

10 Sovereignmemory

Sovereign Memory Benchmark — OPENMODELICA

Company: Anticloud FZ LLE

Performance

Operation Time

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remember() (write+encrypt) 8.2ms

recall() (decrypt+lookup) 3.1ms

search_text (1K entries) 4.8ms

search_semantic (1K entries, 384-dim) 12ms

Reload from disk (10K entries) 340ms

Encryption

Property Value

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Algorithm AES-256-GCM

Key derivation scrypt (N=16384, r=8, p=1)

Nonce 96-bit random per write

Auth tag 128-bit GCM

Password → key time ~200ms (intentionally slow)

Storage

Each entry: ~500 bytes average (compressed JSON + encryption overhead).

10,000 entries ≈ 5MB. Unlimited retention — local disk only.

Data Sovereignty

- Zero cloud sync
- Zero telemetry
- Works fully offline forever
- Password known only to deploying organization